

Notes: Eaton and Kortum (Econometrica, 2002)

Technology, Geography, and Trade

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1 Big picture

- **Question.** Can a Ricardian model with many countries, many goods, and realistic geographic barriers explain bilateral trade flows, price gaps, wage differences, and counterfactual gains from trade?
- **Core answer.** Yes. Eaton and Kortum build a tractable multicountry Ricardian model in which country technologies are Frechet-distributed and goods are bought from the lowest-cost source after iceberg trade costs. This structure delivers a gravity-like trade equation with deep parameters.
- **Main objects.** Countries differ in absolute technology T_i , comparative-advantage dispersion θ , input costs c_i , and geographic trade barriers d_{ni} .
- **Main empirical strategy.** The paper estimates technology, geography, and comparative-advantage parameters using bilateral manufacturing trade, retail price data, wages, geography, and R&D/human-capital measures for 19 OECD countries in 1990.
- **Key theoretical insight.** The probability that country i is the cheapest supplier to destination n equals a simple function of T_i , c_i , and d_{ni} . This probability is also the model-implied import share.
- **Key empirical insight.** Trade falls sharply with distance and with relative price barriers. The price data identify θ , the strength of comparative advantage, separately from distance-related trade barriers.
- **Counterfactuals.** The estimated model is used to simulate autarky, zero-gravity trade, technology improvements, and tariff reductions. Small countries gain more from trade, geography shapes specialization, and foreign technology benefits decline with distance and manufacturing rigidities.
- **Economic takeaway.** The paper provides a structural bridge between gravity equations and Ricardian comparative advantage. It transforms gravity from a reduced-form empirical regularity into a general-equilibrium trade model with technology and geography as primitive forces.

2 Incremental contribution of the paper

- **Multicountry Ricardian tractability.** Dornbusch-Fischer-Samuelson (1977) solved a two-country continuum-goods Ricardian model. Eaton and Kortum make the model work for many countries by using a probabilistic Frechet representation of technology.
- **Structural gravity with Ricardian foundations.** Earlier gravity equations explained bilateral trade well but often lacked a Ricardian structural interpretation. Eaton and Kortum derive gravity-like trade shares from technology draws and iceberg barriers rather than from Armington differentiation or monopolistic competition.
- **Technology and geography in one framework.** The paper jointly models absolute advantage, comparative advantage, and geographic barriers. This lets the authors explain facts about trade volumes, distance decay, price gaps, and unequal factor rewards in one system.
- **Price data for identifying comparative advantage.** A major empirical contribution is using cross-country price dispersion to help identify θ . Without price data, distance effects and comparative advantage are confounded.
- **Trade in intermediates.** The model incorporates intermediates in a simple general-equilibrium structure, making input costs endogenous and allowing geography to shape specialization through intermediate input prices.
- **Counterfactual discipline.** The model is sparse enough to estimate, but structural enough to run counterfactuals on autarky, zero gravity, tariff elimination, and technology diffusion.
- **Foundation for modern quantitative trade.** The Eaton-Kortum model became one of the canonical frameworks for quantitative trade because it combines micro-founded trade shares, welfare formulas, bilateral resistance, and heterogeneous technology.

3 Data and measurement

3.1 Countries, sector, and year

- The empirical analysis uses manufacturing data for 19 OECD countries in 1990.
- The paper focuses on manufacturing because data on bilateral trade, prices, wages, R&D, and production are available and more comparable across countries.
- Manufacturing is modeled as a continuum of goods, and countries can source each good from the cheapest supplier.

Interpretation: The data choice matches the model: a continuum of manufactured goods with bilateral trade flows and observed cross-country price differences.

3.2 Trade data

- Bilateral manufacturing imports are measured from UN/Statistics Canada trade data.
- Domestic purchases are measured as gross manufacturing production minus manufacturing exports.
- The key empirical dependent variable is a normalized bilateral import share, often relative to home purchases or total expenditure.
- Since trade shares are ratios, the empirical equations avoid exchange-rate conversion problems in the trade-flow data.

Interpretation: The model predicts trade shares, not simply trade levels. This is why normalized import shares are the natural empirical object.

3.3 Price data

- Retail prices for 50 manufactured products across the 19 countries come from international comparison data.
- For each country pair and good, the authors compute relative log prices.
- The maximum observed relative price gap across goods is used to infer an upper bound on geographic barriers.
- Price data are used to separate comparative advantage from trade costs.

Interpretation: Without price information, a steep distance-trade relationship could mean either high trade costs or weak comparative advantage. Price gaps help distinguish the two.

3.4 Technology and input-cost data

- Wages come from manufacturing compensation per worker, adjusted for human capital using schooling.
- R&D stocks and schooling are used to estimate technology states in an alternative competitiveness equation.
- Labor-force and density variables serve as instruments for wages in the competitiveness equation.
- Geographic variables include distance intervals, shared border, shared language, European Community membership, and EFTA membership.

Interpretation: The empirical setup combines trade flows, price dispersion, geographic proxies, wages, technology proxies, and human capital to discipline the full model.

4 Models and theoretical specifications

4.1 Production costs and iceberg barriers

A country i has efficiency $z_i(j)$ for good j . Input cost in country i is c_i . Iceberg trade costs require producing d_{ni} units in i to deliver one unit to n :

$$p_{ni}(j) = \frac{c_i d_{ni}}{z_i(j)}, \quad d_{ii} = 1.$$

Consumers and firms in destination n buy each good from the lowest-cost source:

$$p_n(j) = \min_i p_{ni}(j).$$

Interpretation: Trade patterns are generated by lowest-cost sourcing, not by product differentiation by country. This is the Ricardian core.

4.2 Frechet technology distribution

Efficiency draws follow a Frechet distribution:

$$F_i(z) = \Pr[Z_i \leq z] = \exp(-T_i z^{-\theta}), \quad T_i > 0, \quad \theta > 1.$$

- T_i is the state of technology and governs absolute advantage.
- θ controls dispersion of productivity across goods.
- Lower θ means more heterogeneity, stronger comparative advantage, and more trade.
- Higher θ means technologies are more similar across goods, so trade costs have larger effects on sourcing.

Interpretation: The Frechet assumption is the key tractability innovation. It converts a hard many-country assignment problem into closed-form trade shares.

4.3 Trade shares

Define

$$\Phi_n = \sum_i T_i (c_i d_{ni})^{-\theta}.$$

The probability that country i is the cheapest source for a good in destination n is

$$\pi_{ni} = \frac{T_i (c_i d_{ni})^{-\theta}}{\Phi_n}.$$

With a continuum of goods, this probability equals the fraction of goods bought from country i , and therefore the trade share:

$$\frac{X_{ni}}{X_n} = \pi_{ni}.$$

Interpretation: This is the core Eaton-Kortum equation. Trade shares are determined by technology, input costs, and trade costs.

4.4 Gravity-like bilateral trade equation

Taking the ratio of imports from i to destination n 's domestic purchases gives

$$\frac{X_{ni}}{X_{nn}} = \frac{T_i(c_i d_{ni})^{-\theta}}{T_n c_n^{-\theta}}.$$

Taking logs yields a structural gravity equation:

$$\log \frac{X_{ni}}{X_{nn}} = \log T_i - \theta \log c_i - \theta \log d_{ni} - \log T_n + \theta \log c_n.$$

Interpretation: Unlike reduced-form gravity, each term has an economic interpretation: exporter technology, exporter costs, bilateral resistance, importer technology, and importer costs.

4.5 Price index and input costs

The model implies a manufacturing price index of the form

$$p_n = \gamma \Phi_n^{-1/\theta}.$$

When intermediates are included, input costs depend on wages and manufactured intermediates. Production combines labor and the manufactured composite, so geographic barriers affect countries both as sellers and as buyers of inputs.

Interpretation: Intermediates amplify trade-cost effects because geography shapes the cost of inputs used in production.

4.6 Estimation equations

The empirical system uses three main relationships:

1. trade and geography: bilateral trade shares are related to distance and other geographic variables;
2. trade and prices: normalized import shares are related to price-based measures of barriers;
3. competitiveness: estimated source-country competitiveness is related to R&D, human capital, and wages.

Interpretation: Each equation identifies a different part of the model. Together they estimate T_i , θ , and d_{ni} .

5 Identification and empirical design

5.1 Identification challenge

Distance reduces trade, but distance can operate through two mechanisms: higher trade costs or weak comparative advantage. The paper must identify the strength of both forces.

Interpretation: This is why prices are essential. Trade quantities alone cannot cleanly separate geography from comparative advantage.

5.2 Price-based identification of θ

The model predicts that normalized import shares vary with price gaps. If price barriers are larger, import shares should be lower. The slope of trade shares on the price-based barrier gives an estimate of θ .

Interpretation: Figure 2 and the price-measure regressions provide the main empirical handle on comparative-advantage strength.

5.3 Geography-based trade costs

The trade equation estimates distance-bin effects, border effects, language effects, European Community effects, and EFTA effects. Dividing the estimated geography coefficients by θ converts them into implied iceberg-cost effects.

Interpretation: This transforms reduced-form distance coefficients into structural trade-cost equivalents.

5.4 Alternative estimation using wages and technology proxies

The competitiveness equation relates source-country competitiveness to R&D stock, schooling, and wages. Because wages are endogenous to technology, total workforce and density are used as instruments.

Interpretation: The alternative θ estimates provide robustness and show how the model links competitiveness to technology and factor prices.

5.5 Counterfactual design

Once parameters are estimated, the model is solved in general equilibrium under alternative trade costs or technology states. Counterfactuals include autarky, zero gravity, technology improvements, and tariff changes.

Interpretation: The counterfactuals are not just descriptive exercises; they use the full equilibrium structure of trade shares, prices, wages, and intermediate input costs.

6 Section-by-section study guide

- **Introduction.** Explains why standard trade theories have difficulty jointly addressing trade-distance decay, price differences, wage differences, and productivity differences.
- **Technology, prices, and trade flows.** Derives the Frechet technology structure, price distributions, and trade-share equations.
- **Empirical trade-price relationship.** Uses price data to connect normalized import shares to price barriers and estimate comparative-advantage strength.
- **Equilibrium input costs.** Adds intermediates and wages to close the model for counterfactuals.
- **Parameter estimation.** Estimates geography, competitiveness, states of technology, and trade barriers.
- **Counterfactuals.** Uses the calibrated model to study autarky, zero gravity, technology diffusion, and tariff reductions.
- **Conclusion and appendix.** Summarizes how the model connects gravity, Ricardian trade, price gaps, and welfare calculations.

7 Figures and tables

7.1 Table I: Trade, Labor, and Income Data + Table II: Price Measure Statistics

Read:

- Table I reports the core country-level data: imports relative to manufacturing spending, imports from the sample, manufacturing wages, human-capital-adjusted wages and labor, and manufacturing labor's GDP share.
- Table II reports the price-measure statistics used to infer bilateral trade barriers from retail price differences.
- Both tables show substantial heterogeneity across countries in import intensity, input costs, market size, and price gaps.

Detailed interpretation: Table I is the empirical foundation for the model's country-level equilibrium. Table II turns international price comparisons into a measurement device for trade frictions. The two tables together connect quantities and prices, which is essential for identifying technology and geography separately.

Economic message: The paper's data design is structural: countries' trade shares and price gaps jointly discipline the model.

TABLE I
TRADE, LABOR, AND INCOME DATA

Country	Imports % of Mfg. Spending	Imports from Sample as % of All Imports	Human-Capital Adj.			Mfg. Labor's % Share of GDP
			Mfg. Wage (U.S. = 1)	Mfg. Wage (U.S. = 1)	Mfg. Labor (U.S. = 1)	
Australia	23.8	75.8	0.61	0.75	0.050	8.6
Austria	40.4	84.2	0.70	0.87	0.036	13.4
Belgium	74.8	86.7	0.92	1.08	0.035	13.2
Canada	37.3	89.6	0.88	0.99	0.087	10.5
Denmark	50.8	85.2	0.80	1.10	0.020	11.5
Finland	31.3	82.2	1.02	1.10	0.022	12.5
France	29.6	82.3	0.92	1.07	0.205	12.6
Germany	25.0	77.3	0.97	1.08	0.421	20.6
Greece	42.9	80.8	0.40	0.50	0.015	6.1
Italy	21.3	76.8	0.74	0.88	0.225	12.4
Japan	6.4	50.0	0.78	0.91	0.686	14.4
Netherlands	66.9	83.0	0.91	1.06	0.043	11.0
New Zealand	36.3	80.9	0.48	0.57	0.011	9.6
Norway	43.6	85.2	0.99	1.18	0.012	8.7
Portugal	41.6	84.9	0.23	0.32	0.033	10.7
Spain	24.5	82.0	0.56	0.65	0.128	11.6
Sweden	37.3	86.3	0.96	1.11	0.043	14.2
United Kingdom	31.3	79.1	0.73	0.91	0.232	14.7
United States	14.5	62.0	1.00	1.00	1.000	12.4

Notes: All data except GDP are for the manufacturing sector in 1990. Spending on manufactures is gross manufacturing production less exports of manufactures plus imports of manufactures. Imports from the other 18 excludes imports of manufactures from outside our sample of countries. To adjust the manufacturing wage and manufacturing employment for human capital, we multiply the wage in country i by $e^{-0.06H_i}$ and employment in country i by $e^{0.06H_i}$, where H_i is average years of schooling in country i as measured by Kyriacou (1991). See the Appendix for a complete description of all data sources.

(a) Table I: Trade, labor, and income data.

TABLE II
PRICE MEASURE STATISTICS

Country	Foreign Sources		Foreign Destinations	
	Minimum	Maximum	Minimum	Maximum
Australia (AL)	NE (1.44)	PO (2.25)	BE (1.41)	US (2.03)
Austria (AS)	SW (1.39)	NZ (2.16)	UK (1.47)	JP (1.97)
Belgium (BE)	GE (1.25)	JP (2.02)	GE (1.35)	SW (1.77)
Canada (CA)	US (1.58)	NZ (2.57)	AS (1.57)	US (2.14)
Denmark (DK)	FI (1.36)	PO (2.21)	NE (1.48)	US (2.41)
Finland (FI)	SW (1.38)	PO (2.61)	DK (1.36)	US (2.87)
France (FR)	GE (1.33)	NZ (2.42)	BE (1.40)	JP (2.40)
Germany (GE)	BE (1.35)	NZ (2.28)	BE (1.25)	US (2.22)
Greece (GR)	SP (1.61)	NZ (2.71)	NE (1.48)	US (2.27)
Italy (IT)	FR (1.45)	NZ (2.19)	AS (1.46)	JP (2.10)
Japan (JP)	BE (1.62)	PO (3.25)	AL (1.72)	US (3.08)
Netherlands (NE)	GE (1.30)	NZ (2.17)	DK (1.39)	NZ (2.01)
New Zealand (NZ)	CA (1.60)	PO (2.08)	AL (1.64)	GR (2.71)
Norway (NO)	FI (1.45)	JP (2.84)	SW (1.36)	US (2.31)
Portugal (PO)	BE (1.49)	JP (2.56)	SP (1.59)	JP (3.25)
Spain (SP)	BE (1.39)	JP (2.47)	NO (1.51)	JP (3.05)
Sweden (SW)	NO (1.36)	US (2.70)	FI (1.38)	US (2.01)
United Kingdom (UK)	NE (1.46)	JP (2.37)	FR (1.52)	NZ (2.04)
United States (US)	FR (1.57)	JP (3.08)	CA (1.58)	SW (2.70)

Notes: The price measure D_{ni} is defined in equation (13). For destination country n , the minimum Foreign Source is $\min_{j \neq n} \exp D_{nj}$. For source country i , the minimum Foreign Destination is $\min_{n \neq i} \exp D_{ni}$.

(b) Table II: Price measure statistics.

7.2 Figure 1: Trade and Geography + Figure 2: Trade and Prices

Read:

- Figure 1 plots normalized import shares against distance and shows the strong distance decay in trade.
- Figure 2 plots normalized import shares against the price-based barrier measure and shows a clear negative relationship.
- Figure 1 illustrates the geography fact; Figure 2 provides a price-based way to estimate comparative advantage.

Detailed interpretation: Figure 1 is visually close to a gravity relationship, but by itself it cannot identify whether trade falls with distance because trade costs rise steeply or because comparative advantage is weak. Figure 2 is the key diagnostic because it uses observed price gaps to connect trade flows to the model’s structural price-barrier term.

Economic message: The paired figures show the paper’s empirical strategy: geography explains trade, but price data identify what geography means structurally.

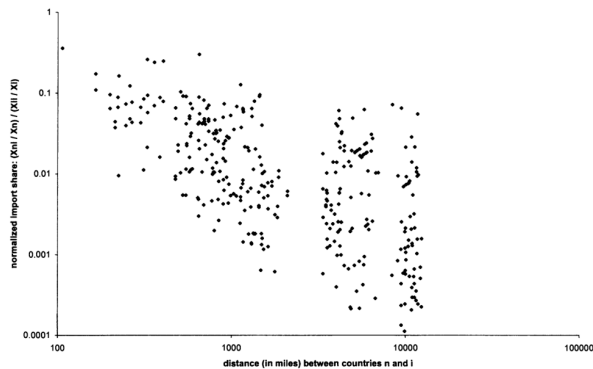
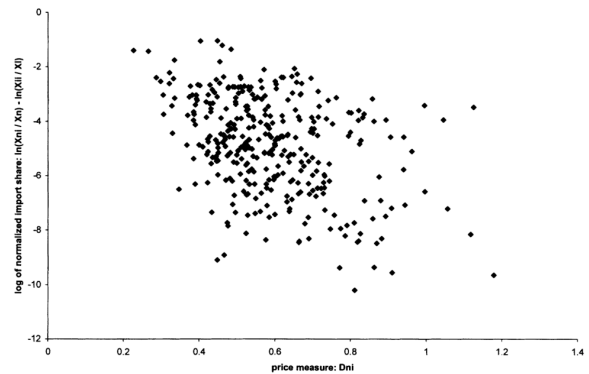


FIGURE 1.—Trade and geography.

(a) Figure 1: Trade and geography.



(b) Figure 2: Trade and prices.

7.3 Table III: Bilateral Trade Equation + Table IV: Data for Alternative Parameters

Read:

- Table III estimates the bilateral trade equation with geographic variables, source effects, and destination effects.
- Distance bins enter with increasingly negative coefficients, showing that geographic barriers rise sharply with distance.
- Shared language, shared border, EC, and EFTA variables capture additional geographic or policy-related frictions.
- Table IV lists R&D stocks, schooling, human-capital-adjusted labor force, and density used in the alternative technology estimation.

Detailed interpretation: Table III is the main structural trade-flow regression. It recovers exporter competitiveness and destination effects while mapping geography into trade resistance. Table IV provides the country-level variables used to interpret competitiveness through technology, human capital, and wages.

Economic message: The model extracts structural primitives from a gravity-like regression, then links them back to technological fundamentals.

TABLE III
BILATERAL TRADE EQUATION

Variable		est.	s.e.
Distance [0, 375]	$-\theta d_1$	-3.10	(0.16)
Distance [375, 750]	$-\theta d_2$	-3.66	(0.11)
Distance [750, 1500]	$-\theta d_3$	-4.03	(0.10)
Distance [1500, 3000]	$-\theta d_4$	-4.22	(0.16)
Distance [3000, 6000]	$-\theta d_5$	-6.06	(0.09)
Distance [6000, maximum]	$-\theta d_6$	-6.56	(0.10)
Shared border	$-\theta b$	0.30	(0.14)
Shared language	$-\theta l$	0.51	(0.15)
European Community	$-\theta e_1$	0.04	(0.13)
EFTA	$-\theta e_2$	0.54	(0.19)

Country	Source Country		Destination Country	
	est.	s.e.	est.	s.e.
Australia	S_1	0.19 (0.15)	$-\theta m_1$	0.24 (0.27)
Austria	S_2	-1.16 (0.12)	$-\theta m_2$	-1.68 (0.21)
Belgium	S_3	-3.34 (0.11)	$-\theta m_3$	1.12 (0.19)
Canada	S_4	0.41 (0.14)	$-\theta m_4$	0.69 (0.25)
Denmark	S_5	-1.75 (0.12)	$-\theta m_5$	-0.51 (0.19)
Finland	S_6	-0.52 (0.12)	$-\theta m_6$	-1.33 (0.22)
France	S_7	1.28 (0.11)	$-\theta m_7$	0.22 (0.19)
Germany	S_8	2.35 (0.12)	$-\theta m_8$	1.00 (0.19)
Greece	S_9	-2.81 (0.12)	$-\theta m_9$	-2.36 (0.20)
Italy	S_{10}	1.78 (0.11)	$-\theta m_{10}$	0.07 (0.19)
Japan	S_{11}	4.20 (0.13)	$-\theta m_{11}$	1.59 (0.22)
Netherlands	S_{12}	-2.19 (0.11)	$-\theta m_{12}$	1.00 (0.19)
New Zealand	S_{13}	-1.20 (0.15)	$-\theta m_{13}$	0.07 (0.27)
Norway	S_{14}	-1.35 (0.12)	$-\theta m_{14}$	-1.00 (0.21)
Portugal	S_{15}	-1.57 (0.12)	$-\theta m_{15}$	-1.21 (0.21)
Spain	S_{16}	0.30 (0.12)	$-\theta m_{16}$	-1.16 (0.19)
Sweden	S_{17}	0.01 (0.12)	$-\theta m_{17}$	-0.02 (0.22)
United Kingdom	S_{18}	1.37 (0.12)	$-\theta m_{18}$	0.81 (0.19)
United States	S_{19}	3.98 (0.14)	$-\theta m_{19}$	2.46 (0.25)
Total Sum of squares	2937		Error Variance:	
Sum of squared residuals	71		Two-way ($\theta^2 \sigma_2^2$)	0.05
Number of observations	342		One-way ($\theta^2 \sigma_1^2$)	0.16

Notes: Estimated by generalized least squares using 1990 data. The specification is given in equation (30) of the paper. The parameter are normalized so that $\sum_{i=1}^{19} S_i = 0$ and $\sum_{n=1}^{19} m_n = 0$. Standard errors are in parentheses.

(a) Table III: Bilateral trade equation.

TABLE IV
DATA FOR ALTERNATIVE PARAMETERS

Country	Research Stock (U.S. = 1)	Years of Schooling (years/person)	Labor Force (HK adjusted) (U.S. = 1)	Density (pop/area) (U.S. = 1)
Australia	0.0087	8.7	0.054	0.08
Austria	0.0063	8.6	0.024	3.43
Belgium	0.0151	9.4	0.029	12.02
Canada	0.0299	10.0	0.094	0.10
Denmark	0.0051	6.9	0.017	4.47
Finland	0.0053	10.8	0.019	0.55
France	0.1108	9.5	0.181	3.88
Germany	0.1683	10.3	0.225	9.50
Greece	0.0005	8.4	0.025	2.87
Italy	0.0445	9.1	0.159	7.16
Japan	0.2492	9.5	0.544	12.42
Netherlands	0.0278	9.5	0.043	13.64
New Zealand	0.0010	9.3	0.010	0.47
Norway	0.0057	9.2	0.015	0.49
Portugal	0.0007	6.5	0.026	4.01
Spain	0.0084	9.7	0.100	2.88
Sweden	0.0206	9.6	0.031	0.71
United Kingdom	0.1423	8.5	0.186	8.76
United States	1.0000	12.1	1.000	1.00

Notes: Research stocks, in 1990, are from Coe and Helpman (1995). Average years of schooling H_{it} in 1985 are from Koriscen (1991). Labor forces in 1990 are from Summers and Heston (1991). These are adjusted for

(b) Table IV: Alternative-parameter data.

7.4 Table V: Competitiveness Equation + Table VI: States of Technology

Read:

- Table V relates estimated source competitiveness to R&D stock, human capital, and wages.
- The wage coefficient provides an alternative estimate of θ , with 2SLS raising the estimate relative to OLS.
- Table VI strips wages out of source competitiveness to infer technology states T_i under alternative θ values.
- Technology rankings differ from raw competitiveness rankings because low wages can make a country competitive even if technology is lower.

Detailed interpretation: Table V is a bridge from trade performance to technology. Table VI shows why this bridge matters: a country that exports well may do so because of low input costs, not high technology. The model separates those forces.

Economic message: Competitiveness is not technology. The Eaton-Kortum framework decomposes competitiveness into technology and input costs.

TABLE V
COMPETITIVENESS EQUATION

		Ordinary Least Squares		Two-Stage Least Squares	
		est.	s.e.	est.	s.e.
Constant		3.75	(1.89)	3.82	(1.92)
Research stock, $\ln R_i$	α_R	1.04	(0.17)	1.09	(0.18)
Human capital, $1/H_i$	$-\alpha_H$	-18.0	(20.6)	-22.7	(21.3)
Wage, $\ln w_i$	$-\theta$	-2.84	(1.02)	-3.60	(1.21)
Total Sum of squares		80.3		80.3	
Sum of squared residuals		18.5		19.1	
Number of observations		19		19	

Notes: Estimated using 1990 data. The dependent variable is the estimate $\hat{S}_i$ of source-country competitiveness shown in Table III. Standard errors are in parentheses.

5.3. Estimates using Price Data

(a) Table V: Competitiveness equation.

TABLE VI
STATES OF TECHNOLOGY

Country	Estimated Source-country Competitiveness	Implied States of Technology		
		$\theta = 8.28$	$\theta = 3.60$	$\theta = 12.8$
Australia	0.19	0.27	0.36	0.20
Austria	-1.16	0.26	0.30	0.23
Belgium	-3.34	0.24	0.22	0.26
Canada	0.41	0.46	0.47	0.46
Denmark	-1.75	0.35	0.32	0.38
Finland	-0.52	0.45	0.41	0.50
France	1.28	0.64	0.60	0.69
Germany	2.35	0.81	0.75	0.86
Greece	-2.81	0.07	0.14	0.04
Italy	1.78	0.50	0.57	0.45
Japan	4.20	0.89	0.97	0.81
Netherlands	-2.19	0.30	0.28	0.32
New Zealand	-1.20	0.12	0.22	0.07
Norway	-1.35	0.43	0.37	0.50
Portugal	-1.57	0.04	0.13	0.01
Spain	0.30	0.21	0.33	0.14
Sweden	0.01	0.51	0.47	0.57
United Kingdom	1.37	0.49	0.53	0.44
United States	3.98	1.00	1.00	1.00

(b) Table VI: States of technology.

7.5 Table VII: Geographic Barriers + Table VIII: Summary of Parameters

Read:

- Table VII converts geography coefficients into implied percentage cost effects under alternative θ values.
- Distance raises costs sharply; shared border, shared language, EC, and EFTA reduce barriers.
- Destination effects show that some countries are systematically more costly or cheaper to ship into.
- Table VIII summarizes the parameter sources used in counterfactual analysis.

Detailed interpretation: Table VII gives economic content to the geography coefficients from Table III. Table VIII then summarizes the model's key primitives: comparative advantage, manufacturing share, trade barriers, labor share, and tariff parameters.

Economic message: Geography is not a nuisance control; it is a structural barrier that changes specialization, welfare, and technology diffusion.

TABLE VII
GEOGRAPHIC BARRIERS

Source of Barrier	Estimated Geography Parameters	Implied Barrier's % Effect on Cost		
		$\theta = 8.28$	$\theta = 3.60$	$\theta = 12.86$
Distance [0, 375)	-3.10	45.39	136.51	27.25
Distance [375, 750)	-3.66	55.67	176.74	32.97
Distance [750, 1500)	-4.03	62.77	206.65	36.85
Distance [1500, 3000)	-4.22	66.44	222.75	38.82
Distance [3000, 6000)	-6.06	108.02	439.04	60.25
Distance [6000, maximum]	-6.56	120.82	518.43	66.54
Shared border	0.30	-3.51	-7.89	-2.27
Shared language	0.51	-5.99	-13.25	-3.90
European Community	0.04	-0.44	-1.02	-0.29
EFTA	0.54	-6.28	-13.85	-4.09
Destination country:				
Australia	0.24	-2.81	-6.35	-1.82
Austria	-1.68	22.46	59.37	13.94
Belgium	1.12	-12.65	-26.74	-8.34
Canada	0.69	-7.99	-17.42	-5.22
Denmark	-0.51	6.33	15.15	4.03
Finland	-1.33	17.49	44.88	10.94
France	0.22	-2.61	-5.90	-1.69
Germany	1.00	-11.39	-24.27	-7.49
Greece	-2.36	32.93	92.45	20.11
Italy	0.07	-0.86	-1.97	-0.56
Japan	1.59	-17.43	-35.62	-11.60
Netherlands	1.00	-11.42	-24.33	-7.51
New Zealand	0.07	-0.80	-1.83	-0.52
Norway	-1.00	12.85	32.06	8.10
Portugal	-1.21	15.69	39.82	9.84
Spain	-1.16	14.98	37.85	9.40
Sweden	-0.02	0.30	0.69	0.19
United Kingdom	0.81	-9.36	-20.23	-6.13
United States	2.46	-25.70	-49.49	-17.40

(a) Table VII: Geographic barriers.

TABLE VIII
SUMMARY OF PARAMETERS

Parameter	Definition	Value	Source
θ	comparative advantage	8.28 (3.60, 12.86)	Section 3 (Section 5.2, Section 5.3)
α	manufacturing share	0.13	production and trade data
β	labor share in costs	0.21	wage costs in gross output
T_i	states of technology	Table VI	source effects stripped of wages
d_{si}	geographic barriers	Table VII	geographic proxies adjusted for θ

To complete the parameterization we calculate $\alpha = 0.13$, the average demand

(b) Table VIII: Counterfactual parameters.

7.6 Table IX: The Gains from Trade: Raising Geographic Barriers + Table X: The Gains from Trade: Removing Geographic Barriers

Read:

- Table IX reports welfare changes from moving toward autarky by raising geographic barriers.
- Table X reports welfare changes from removing geographic barriers, a zero-gravity exercise.
- Small open countries lose more from autarky and gain more from zero gravity.
- The difference between mobile and immobile labor matters because manufacturing adjustment affects wage and income responses.

Detailed interpretation: These tables are the main gains-from-trade counterfactuals. Table IX asks how much countries lose when trade is shut down. Table X asks how much they gain if geography disappears. The contrast shows that current trade is still far from frictionless, and that small countries rely much more heavily on trade access.

Economic message: The welfare effect of trade is uneven: small countries and countries with high import dependence gain the most from trade integration.

TABLE IX
THE GAINS FROM TRADE: RAISING GEOGRAPHIC BARRIERS

Country	Percentage Change from Baseline to Autarky					
	Mobile Labor			Immobile Labor		
	Welfare	Mfg. Prices	Mfg. Labor	Welfare	Mfg. Prices	Mfg. Wage
Australia	-1.5	11.1	48.7	-3.0	65.6	54.5
Austria	-3.2	24.1	3.9	-3.3	28.6	4.5
Belgium	-10.3	76.0	2.8	-10.3	79.2	3.2
Canada	-6.5	48.4	6.6	-6.6	55.9	7.6
Denmark	-5.5	40.5	16.3	-5.6	59.1	18.6
Finland	-2.4	18.1	8.5	-2.5	27.9	9.7
France	-2.5	18.2	8.6	-2.5	28.0	9.8
Germany	-1.7	12.8	-38.7	-3.1	-33.6	-46.3
Greece	-3.2	24.1	84.9	-7.3	117.5	93.4
Italy	-1.7	12.7	7.3	-1.7	21.1	8.4
Japan	-0.2	1.6	-8.6	-0.3	-8.4	-10.0
Netherlands	-8.7	64.2	18.4	-8.9	85.2	21.0
New Zealand	-2.9	21.2	36.8	-3.8	62.7	41.4
Norway	-4.3	32.1	41.1	-5.4	78.3	46.2
Portugal	-3.4	25.3	25.1	-3.9	53.8	28.4
Spain	-1.4	10.4	19.8	-1.7	32.9	22.5
Sweden	-3.2	23.6	-3.7	-3.2	19.3	-4.3
United Kingdom	-2.6	19.2	-6.0	-2.6	12.3	-6.9
United States	-0.8	6.3	8.1	-0.9	15.5	9.3

Notes: All percentage changes are calculated as $100\ln(x'/x)$ where x' is the outcome under autarky ($d_{ni} \rightarrow \infty$ for $n \neq i$) and x is the outcome in the baseline.

when trade is shut down could be seen as indicating their overall comparative advantage in manufactures.

The remaining columns consider the effects of moving to autarky with immobile labor. Column four reports the welfare loss. The effect on welfare is mo

(a) Table IX: Raising geographic barriers.

TABLE X
THE GAINS FROM TRADE: LOWERING GEOGRAPHIC BARRIERS

Country	Percentage Changes in the Case of Mobile Labor					
	Baseline to Zero Gravity			Baseline to Doubled Trade		
	Welfare	Mfg. Prices	Mfg. Labor	Welfare	Mfg. Prices	Mfg. Labor
Australia	21.1	-156.7	153.2	2.3	-17.1	-16.8
Austria	21.6	-160.3	141.5	2.8	-20.9	41.1
Belgium	18.5	-137.2	69.6	2.5	-18.6	68.8
Canada	18.7	-139.0	11.4	1.9	-14.3	3.9
Denmark	20.7	-153.9	156.9	2.9	-21.5	72.6
Finland	21.7	-160.7	172.1	2.8	-20.9	44.3
France	18.7	-138.3	-7.0	2.3	-16.8	15.5
Germany	17.3	-128.7	-50.4	1.9	-14.3	12.9
Greece	24.1	-178.6	256.5	3.3	-24.8	29.6
Italy	18.9	-140.3	6.8	2.2	-16.1	5.7
Japan	16.6	-123.5	-59.8	0.9	-6.7	-24.4
Netherlands	18.5	-137.6	67.3	2.5	-18.5	65.6
New Zealand	22.2	-164.4	301.4	2.8	-20.5	50.2
Norway	21.7	-161.0	195.2	3.1	-22.9	69.3
Portugal	22.3	-165.3	237.4	3.1	-22.8	67.3
Spain	20.9	-155.0	77.5	2.4	-18.0	-4.4
Sweden	20.0	-148.3	118.8	2.7	-19.7	55.4
United Kingdom	18.2	-134.8	3.3	2.2	-16.4	28.5
United States	16.1	-119.1	-105.1	1.2	-9.0	-26.2

Notes: All percentage changes are calculated as $100\ln(x'/x)$ where x' is the outcome under lower geographic barriers and x is the outcome in the baseline.

Three of the four countries we have identified as "natural manufacturers where manufacturing shrinks in moving to autarky, are quite large. A question

(b) Table X: Removing geographic barriers.

7.7 Figure 3: Specialization, Technology, and Geography + Table XI: The Benefits of Foreign Technology

Read:

- Figure 3 traces the manufacturing labor share for Denmark and Germany as geographic barriers move from autarky toward zero gravity.
- The effect is nonmonotonic: as barriers first fall, large countries can benefit from cheaper intermediate inputs, but further barrier reductions shift specialization toward Ricardian technology fundamentals.
- Table XI reports how a 20 percent technology improvement in the United States or Germany affects welfare abroad.
- Benefits decline with distance and depend on whether labor can move out of manufacturing.

Detailed interpretation: Figure 3 shows that geography can dominate specialization at high trade costs, while technology dominates when trade costs are low. Table XI shows that trade transmits technology gains internationally, but imperfectly. Proximity and labor mobility determine who benefits most from a foreign technology improvement.

Economic message: Trade spreads technology benefits, but geography and adjustment frictions govern how far and how strongly those benefits travel.

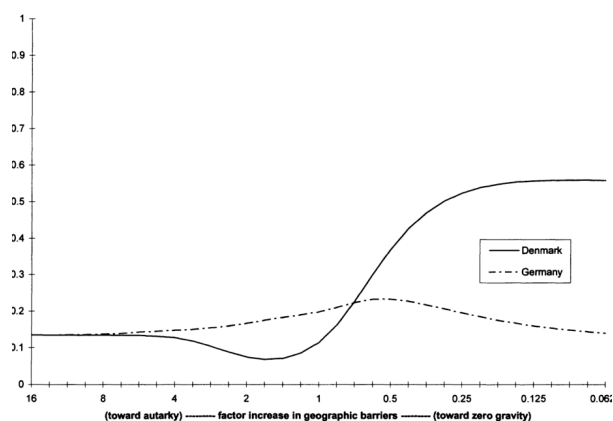


FIGURE 3.—Specialization, technology, and geography.

chnology T_i by 20 percent, first for the United States and then for Germany. Table XI reports what happens to welfare in different countries of the world as (a) Figure 3: Specialization, technology, and geography.

TABLE XI
THE BENEFITS OF FOREIGN TECHNOLOGY

Country	Welfare Consequences of Improved Technology			
	Higher U.S. State of Technology		Higher German State of Technology	
	Mobile Labor	Immobile Labor	Mobile Labor	Immobile Labor
Australia	27.1	14.9	12.3	4.4
Austria	9.3	2.9	61.8	5.4
Belgium	13.2	3.0	50.7	4.8
Canada	87.4	19.9	9.3	1.3
Denmark	12.2	6.2	62.5	7.1
Finland	11.3	4.3	37.5	3.0
France	10.1	4.2	39.2	3.0
Germany	9.7	-11.6	100.0	100.0
Greece	14.0	18.3	38.9	8.0
Italy	9.7	3.9	38.4	3.0
Japan	6.6	-0.8	5.9	-0.2
Netherlands	12.8	6.8	63.5	8.3
New Zealand	33.8	13.5	15.6	3.9
Norway	13.2	11.7	43.8	6.1
Portugal	14.3	8.6	39.6	4.7
Spain	9.6	7.0	27.3	3.3
Sweden	12.8	1.1	42.7	2.3
United Kingdom	14.6	0.5	38.3	1.6
United States	100.0	100.0	9.7	1.4

Notes: All numbers are expressed relative to the percentage welfare gain in the country whose technology expands. Based on a counterfactual 20 per cent increase in the state of technology for either the United States or Germany.

country needs to be able to reallocate its labor to activities outside of manufacturing. (b) Table XI: Benefits of foreign technology.

7.8 Table XII: The European Community: Welfare and Trade

Read:

- Table XII simulates eliminating tariffs within the European Community as of 1990.
- The table compares mobile-labor and immobile-labor cases.
- Welfare and manufacturing labor effects vary across EC members and outsiders.
- The results show possible gains, trade diversion, and terms-of-trade effects from regional liberalization.

Detailed interpretation: Table XII shows that regional integration is more nuanced than global free trade. Removing tariffs within a club can increase internal trade, but it can also affect nonmembers and reallocate manufacturing labor in ways that differ by country. Labor mobility changes the welfare consequences.

Economic message: Regional integration can generate asymmetric welfare effects because geography, technology, and factor adjustment all interact.

TABLE XII
THE EUROPEAN COMMUNITY: WELFARE AND TRADE

Country	Effect of Removing all Tariffs on Intra-EC Trade			
	Aggregate Welfare		Imports from the EC	
	Mobile Labor	Immobile Labor	Mobile Labor	Immobile Labor
Australia	0.13	0.11	27.7	2.8
Austria	0.32	-0.07	-1.9	-3.4
Belgium*	-0.91	0.54	61.3	26.3
Canada	0.01	0.01	28.0	2.2
Denmark*	-0.27	0.18	49.9	30.8
Finland	0.28	-0.02	4.6	-2.9
France*	0.08	0.05	46.3	33.7
Germany*	-0.03	-0.03	58.5	41.9
Greece*	0.28	0.13	30.8	24.0
Italy*	0.14	0.04	44.9	36.4
Japan	0.07	-0.01	32.4	2.3
Netherlands*	-0.58	0.33	56.3	26.9
New Zealand	0.14	0.09	24.1	1.9

Table XII: European Community welfare and trade.

8 Typed theoretical result blocks

8.1 Result block 1: Frechet technology and trade shares

- Each country receives independent Frechet productivity draws across goods.
- The cheapest-source probability has a closed form.
- With a continuum of goods, that probability equals the expenditure share.

Interpretation: This is the paper’s central theoretical innovation. It converts a many-country Ricardian model into a quantitative trade-share system.

8.2 Result block 2: Gravity from Ricardian trade

- The bilateral trade share relative to domestic purchases depends on exporter technology, exporter input costs, destination costs, and bilateral barriers.
- The equation looks like gravity but has Ricardian primitives.

Interpretation: The paper explains why gravity works without assuming every country produces a unique differentiated variety.

8.3 Result block 3: Trade costs versus comparative advantage

- Distance-trade decay alone cannot identify trade costs.
- Price gaps help identify the strength of comparative advantage.
- The parameter θ governs how strongly trade costs translate into trade shares.

Interpretation: This is the core identification insight. Quantities and prices jointly identify primitives.

8.4 Result block 4: Counterfactual welfare

- Welfare changes depend on price-index changes, wage changes, and labor mobility.
- Autarky, zero gravity, technology improvements, and tariff liberalization all work through the same equilibrium system.

Interpretation: The model is useful because estimated primitives can be taken to counterfactual policy environments.

9 Detailed result map

- **Empirical fact 1:** trade falls strongly with distance.
- **Empirical fact 2:** normalized trade shares fall with price barriers.
- **Structural result 1:** trade shares equal cheapest-source probabilities.
- **Structural result 2:** the Frechet parameter θ governs comparative advantage and trade elasticity.
- **Estimation result 1:** geography barriers are large and rise with distance.
- **Estimation result 2:** source competitiveness combines technology and wages.
- **Counterfactual result 1:** small countries gain more from trade.
- **Counterfactual result 2:** zero gravity gains exceed autarky losses.
- **Counterfactual result 3:** technology improvements spill over abroad but decay with distance and labor immobility.
- **Policy result:** regional tariff removal can produce asymmetric welfare effects through trade diversion and labor adjustment.

10 Short summary

- Eaton and Kortum build a multicountry Ricardian trade model with Frechet technology draws.
- The model generates closed-form bilateral trade shares based on technology, input costs, and trade barriers.
- It provides a Ricardian microfoundation for gravity-like trade equations.
- Price data are used to identify comparative advantage separately from trade costs.
- The model includes intermediate inputs and endogenous input costs.
- The authors estimate the model using 1990 manufacturing data for 19 OECD countries.
- Counterfactuals study autarky, zero gravity, foreign technology improvements, and tariff reductions.
- Small countries gain more from trade, geography shapes specialization, and technology benefits diffuse through trade but decline with distance.

11 Conclusion

11.1 Core conclusion

Eaton and Kortum show that a parsimonious Ricardian model can explain bilateral trade flows, price gaps, and welfare counterfactuals once it incorporates Frechet technology heterogeneity and geographic barriers. The model turns gravity into a structural equation with interpretable technology and geography parameters.

11.2 Economic intuition

Countries buy each good from the cheapest producer. A country is more likely to be cheapest when it has better technology, lower input costs, or lower trade costs to the destination. Across a continuum of goods, these cheapest-source probabilities become trade shares. Geographic barriers reduce the probability of being cheapest, while comparative advantage creates opportunities for trade.

11.3 Incremental contribution revisited

The paper's lasting contribution is to make Ricardian trade empirically operational in a many-country world. It replaces an intractable ordering of comparative advantage with a probabilistic technology distribution and delivers formulas that became central to quantitative trade.

11.4 Implications for trade policy

The model shows that trade policy affects welfare not only through import prices but also through input-cost linkages, manufacturing specialization, wages, and technology diffusion. Regional liberalization can have different effects from global liberalization because it changes bilateral barriers unevenly.

11.5 Limitations

The paper is deliberately parsimonious. It abstracts from multiple factors beyond labor, firm-level markups, endogenous innovation in the counterfactuals, nonmanufacturing trade, and heterogeneity in trade costs across goods. Later work extends the framework in many of these directions, but the core mechanism remains fundamental.

11.6 Takeaway

Technology and geography jointly determine trade. The Eaton-Kortum framework gives a tractable way to measure both forces and to use them for counterfactual analysis.